

Pysteps in Operations

Some background to nowcasting and how it can be used



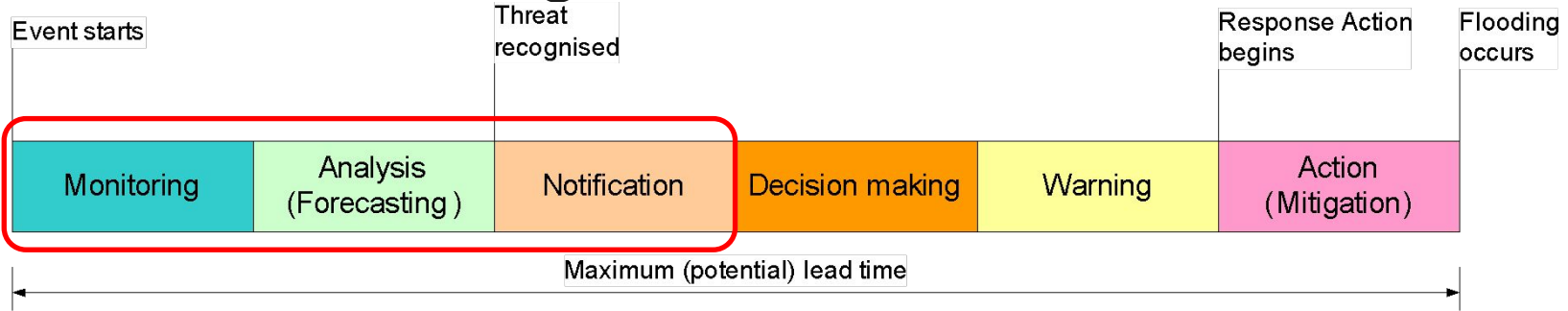
Session overview

Time	Content	Speaker	Duration
09:00	Welcome / Course overview	Ruben	20'
09:20	Lecture: "Pysteps in operations"	Ruben, Lesley	40'
10:00	Coffee break		30'
10:30	Hands-on session "Starting with pysteps"	Jenna, Miguel, Mats	120'
12:30	Lunch break		60'
13:30	Hands-on session "Blending with NWP"	Mats, Lesley, Ruben	120'
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16:00	Lecture: "AI for nowcasting"	Lesley, Jenna, Ruben	45'
16:45	Discussion	All	15'
17:00	End of the short course		

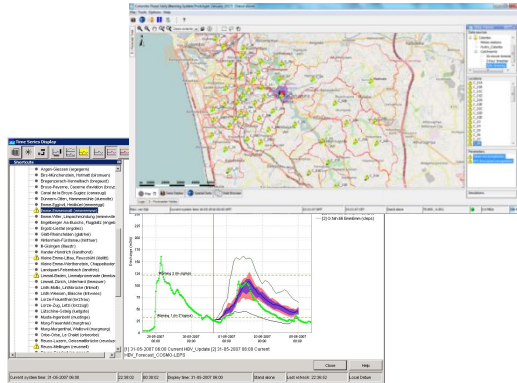


Why we need improved rainfall forecasts for short lead times

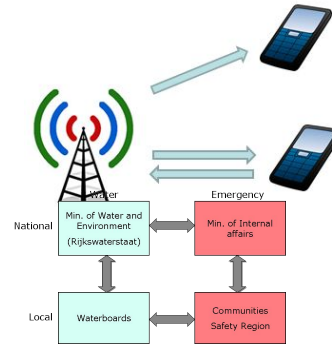
Forecasts, warning and action



Forecast



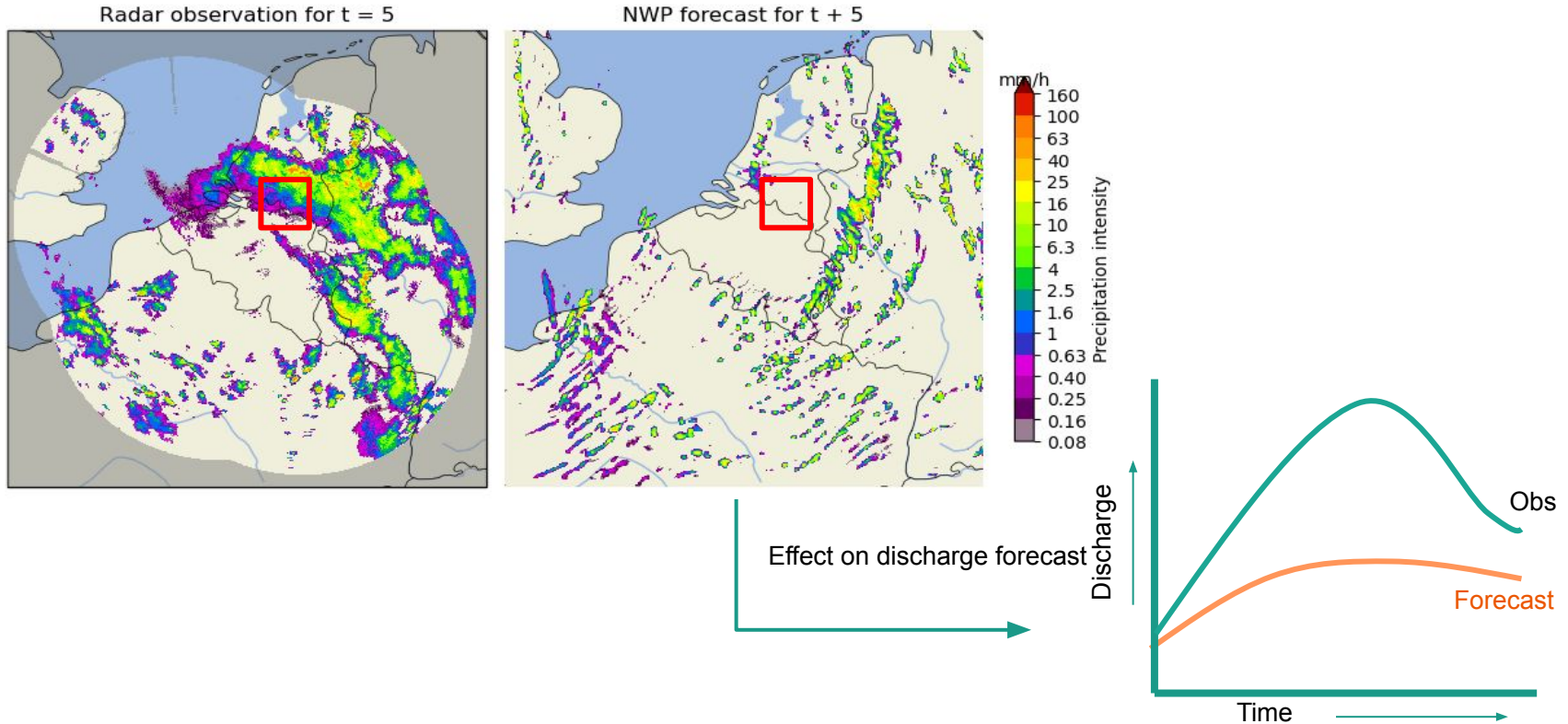
Warning



Action



The limitation of NWP models on short lead times for flood early warning



What is nowcasting and why should we use it?

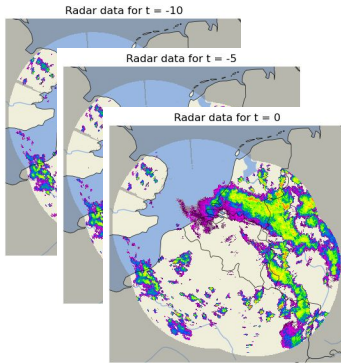


How do we make a nowcast?

Extrapolation

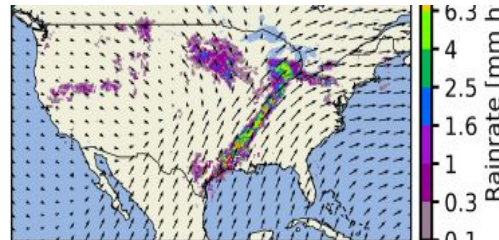
Fast!

But no physics involved..



From most recent weather
radar observations

Determine
movement of
rainfall fields



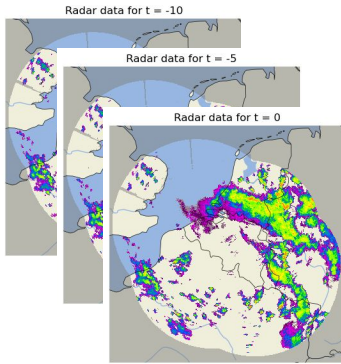
Source: Pulkkinen et al., Geosci. Model. Dev., 2019

How do we make a nowcast?

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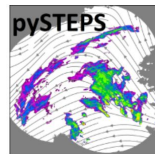
From most recent weather
radar observations

Determine
movement of
rainfall fields

Rainfall field
evolution



Source: Pulkkinen et al., Geosci. Model. Dev., 2019

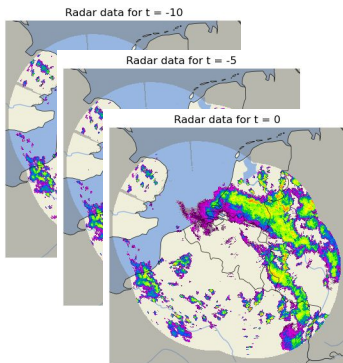


How do we make a nowcast?

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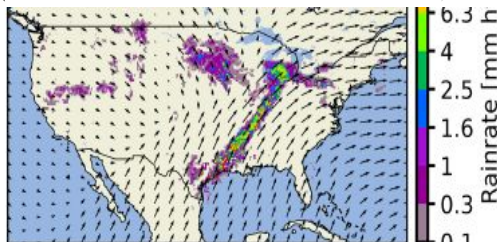
From most recent weather radar observations

Determine movement of rainfall fields

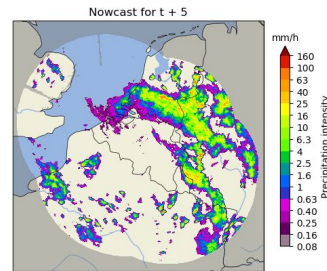
Rainfall field evolution

Add stochastic noise

Extrapolate to future



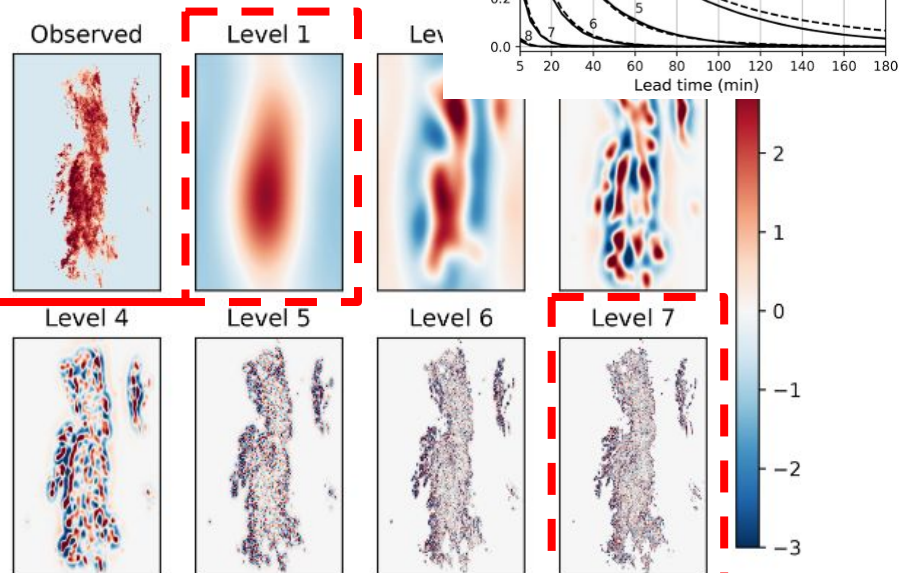
Source: Pulkkinen et al., Geosci. Model. Dev., 2019



To rainfall forecast up to max. 6 hr ahead

The STEPS algorithm in pysteps

- Past research has shown that the lifetime of rainfall fields relates to its spatial scale (e.g., Venugopal et al., 1999, j. Geophys. Res.-Atmos.)
- STEPS takes this into account by decomposing the rainfall fields in spatial cascades with a fast Fourier transform.

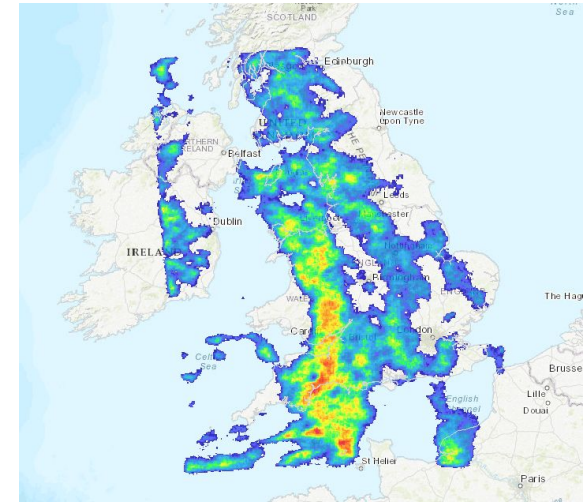
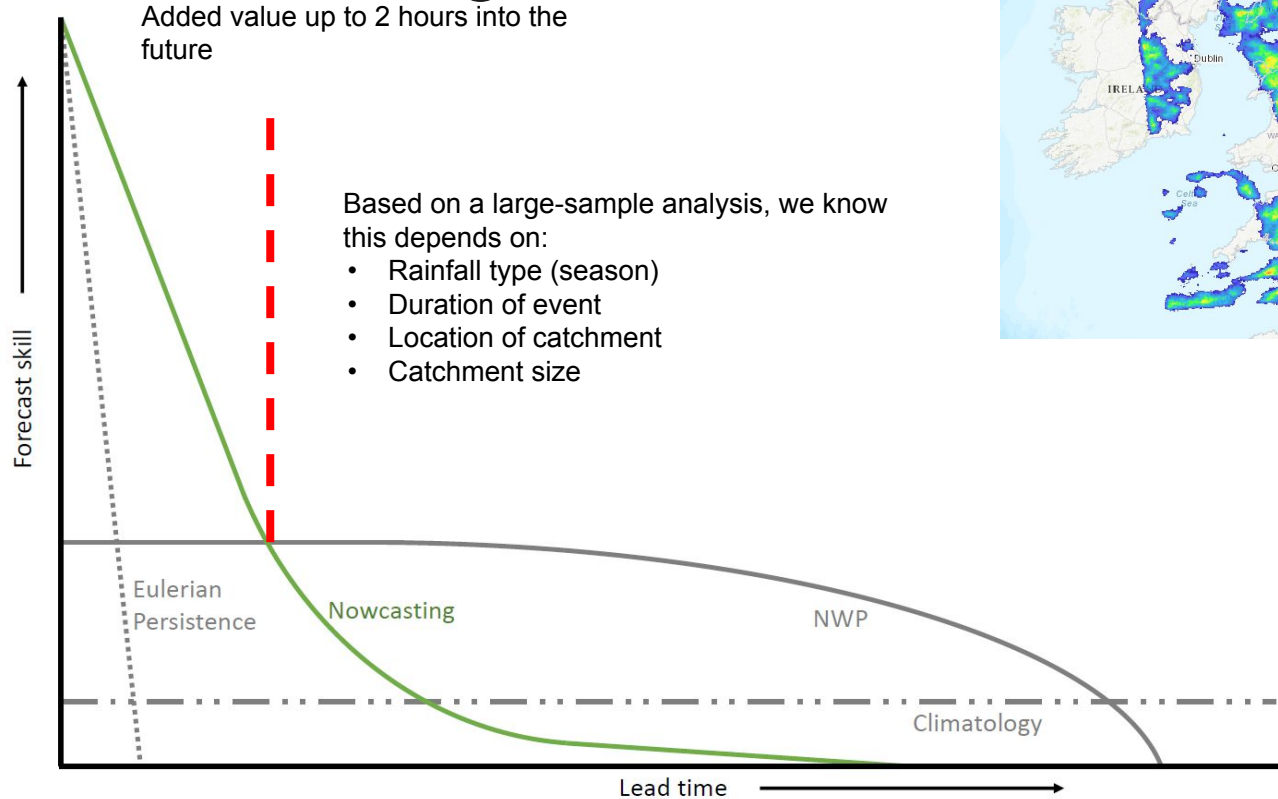


Largest spatial scale: **long lifetime**, advects (moves) without much growth and dissipation

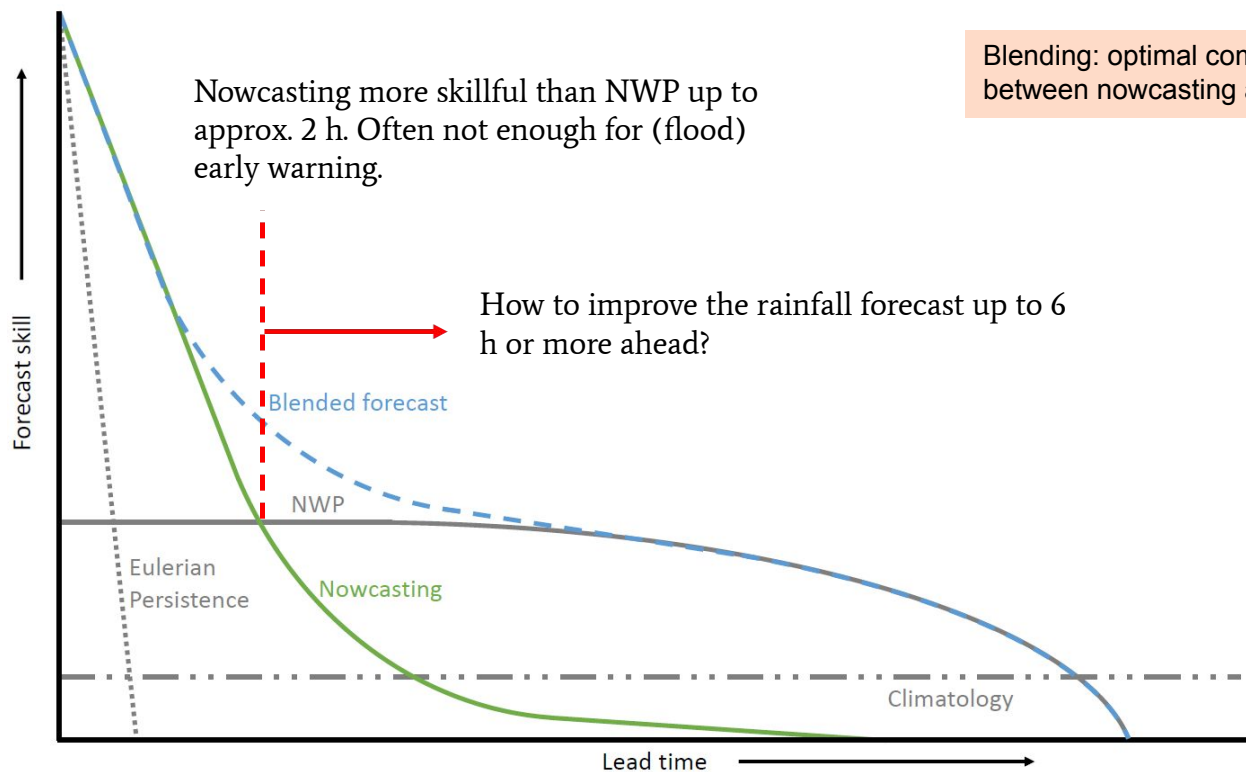
Smallest spatial scale: **short lifetime**, advects (moves) and changes with a combination of **autoregression** and **stochastic perturbations (to model growth and dissipation)** using spatially correlated Gaussian noise.

Source: Pulkkinen et al., 2019, GMD

The skill of nowcasting



Can we even go a step further?



Blending: optimal combination between nowcasting and NWP

Nowcasting more skillful than NWP up to approx. 2 h. Often not enough for (flood) early warning.

How to improve the rainfall forecast up to 6 h or more ahead?

Blended forecast

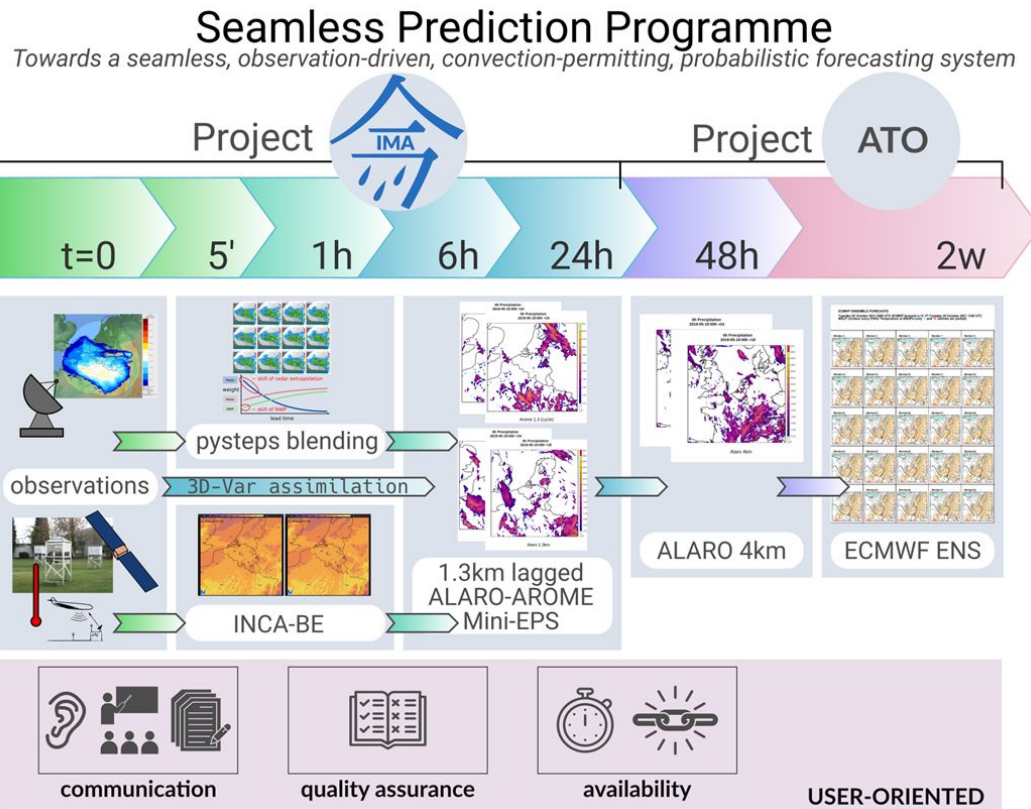
NWP

Eulerian
Persistence

Nowcasting

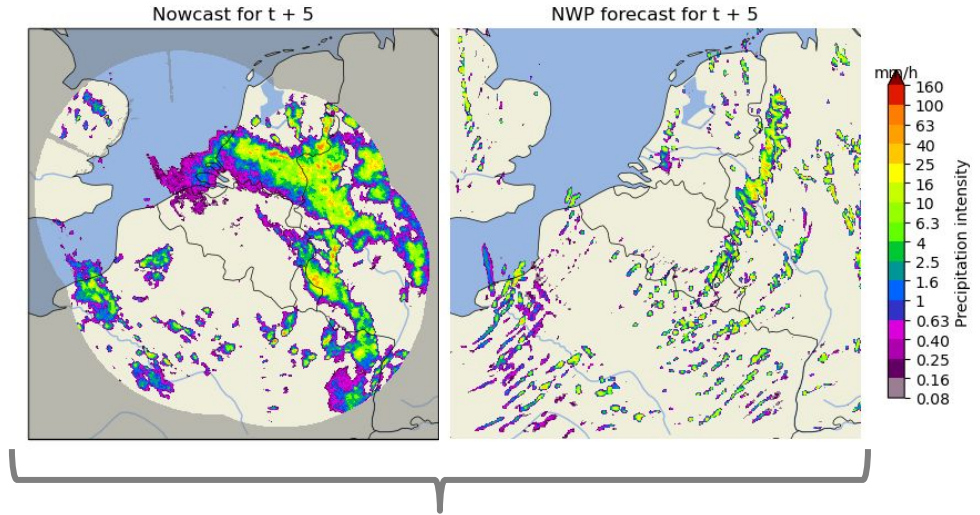
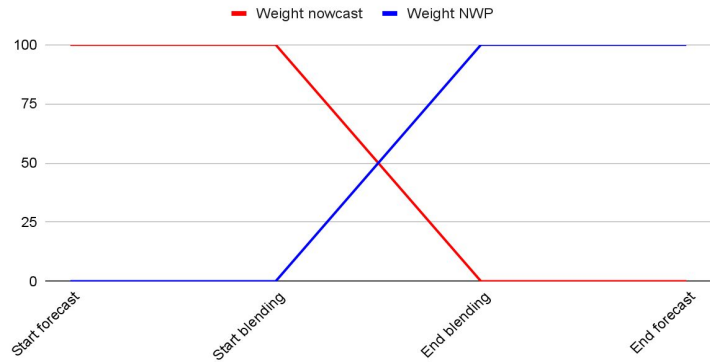
Climatology

Seamless forecasting



Simple linear blending?

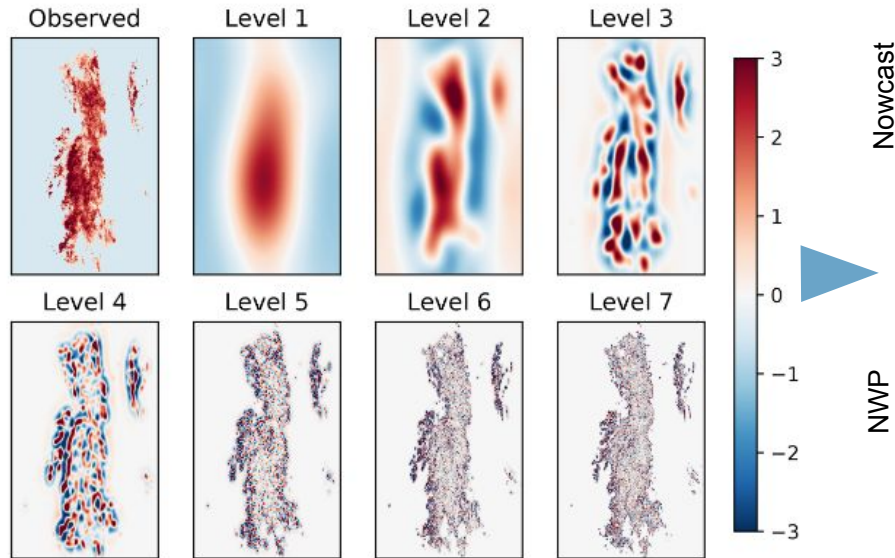
Linear blending weights



How to combine
a case like this?

An advanced blending method (STEPS) – included in the open-source pysteps framework

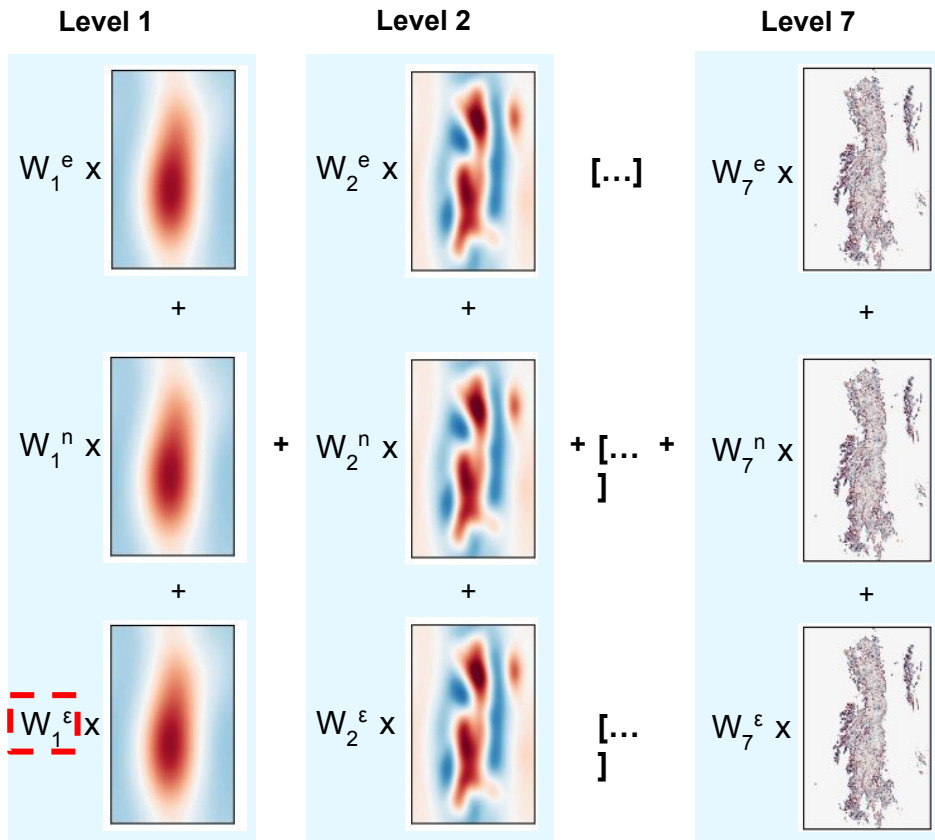
Blending per spatial scale:



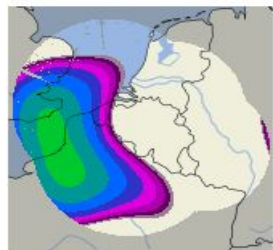
Weights depend on initial skill and expected future skill

Stochastic perturbation

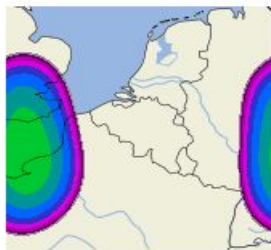
Per ensemble member:



NWC cascade level 0
2022-05-20 08:00 UTC + 5 min



NWP cascade level 0
2022-05-20 00:00 UTC + 485 min



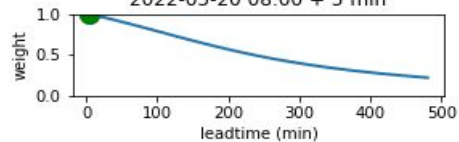
Noise cascade level 0
2022-05-20 08:00 UTC + 5 min



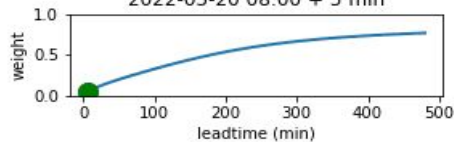
Blended cascade level 0
2022-05-20 08:00 UTC + 5 min



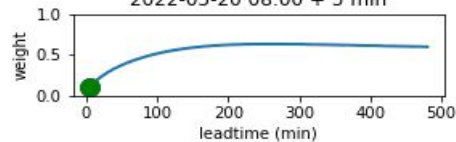
NWC weight cascade level 0
2022-05-20 08:00 + 5 min



NWP weight cascade level 0
2022-05-20 08:00 + 5 min



Noise weight cascade level 0
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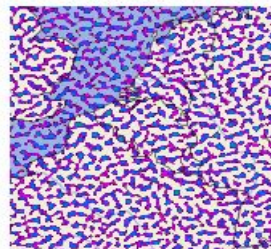
NWC cascade level 4
2022-05-20 08:00 UTC + 5 min



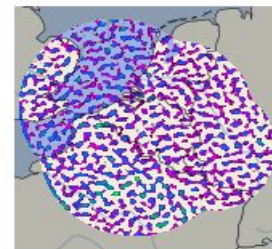
NWP cascade level 4
2022-05-20 00:00 UTC + 485 min



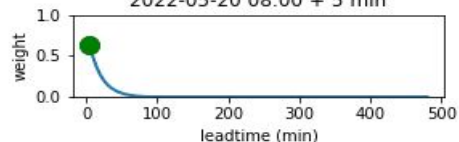
Noise cascade level 4
2022-05-20 08:00 UTC + 5 min



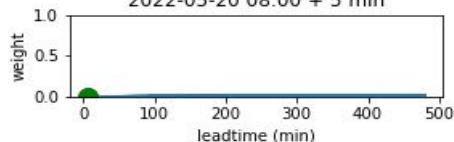
Blended cascade level 4
2022-05-20 08:00 UTC + 5 min



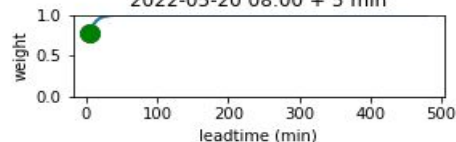
NWC weight cascade level 4
2022-05-20 08:00 + 5 min



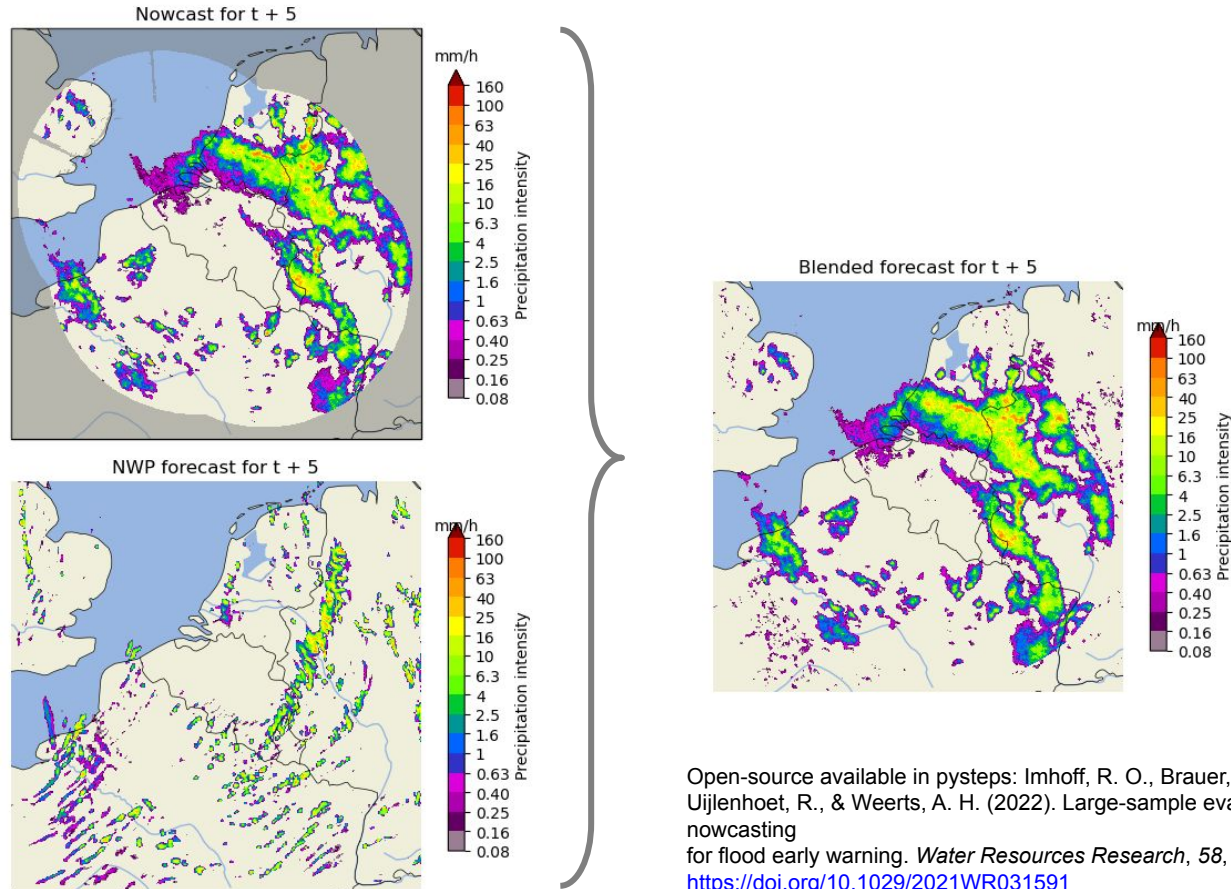
NWP weight cascade level 4
2022-05-20 08:00 + 5 min



Noise weight cascade level 4
2022-05-20 08:00 + 5 min

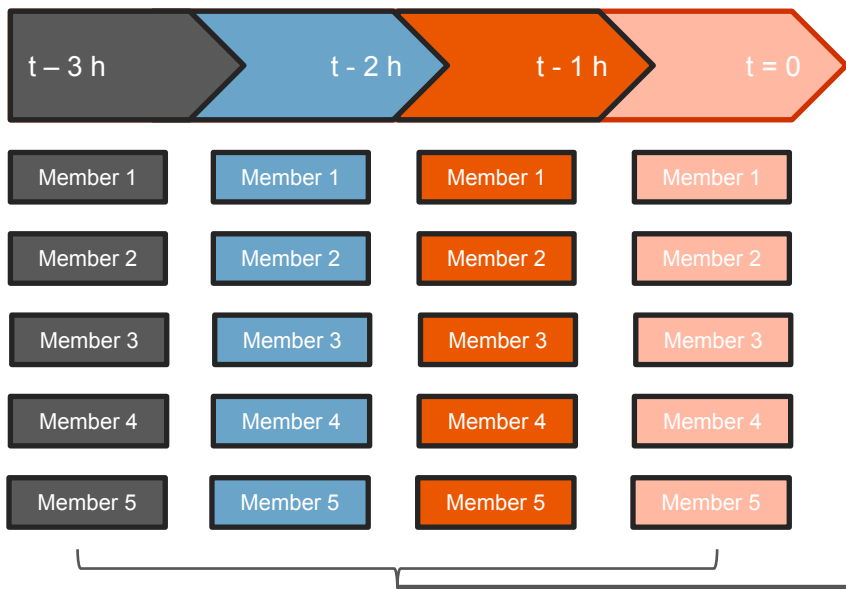


An advanced blending method



Open-source available in pysteps: Imhoff, R. O., Brauer, C. C., van Heeringen, K. J., Uijlenhoet, R., & Weerts, A. H. (2022). Large-sample evaluation of radar rainfall nowcasting for flood early warning. *Water Resources Research*, 58, e2021WR031591. <https://doi.org/10.1029/2021WR031591>

Going seamless: Operationalization of pysteps at KNMI



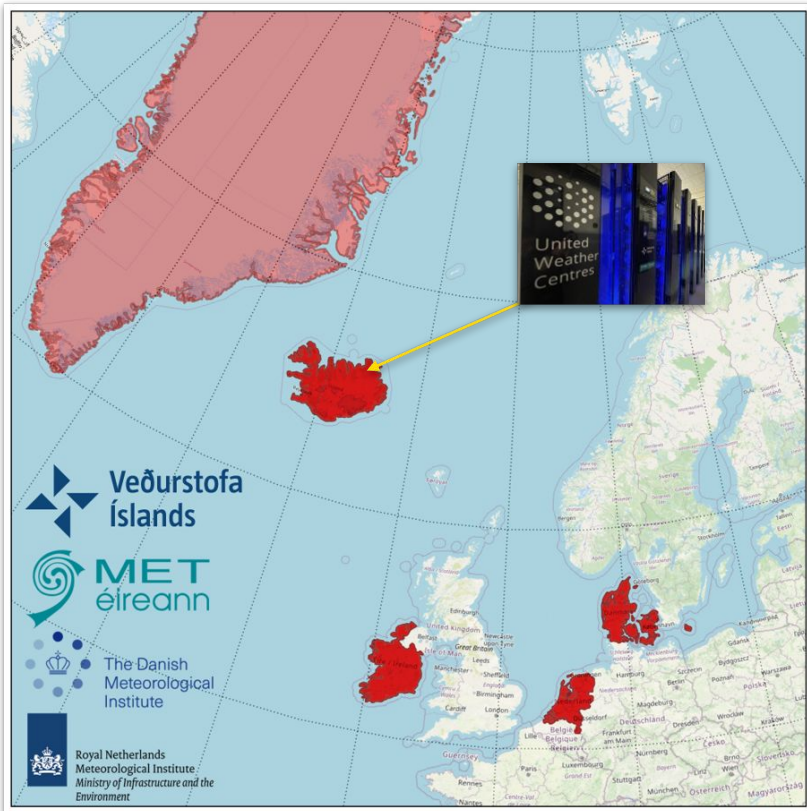
Deterministic nowcast:

- Issued every 5 min.
- Up to 2 hours ahead.
- Only nowcast for visualization in app, etc.

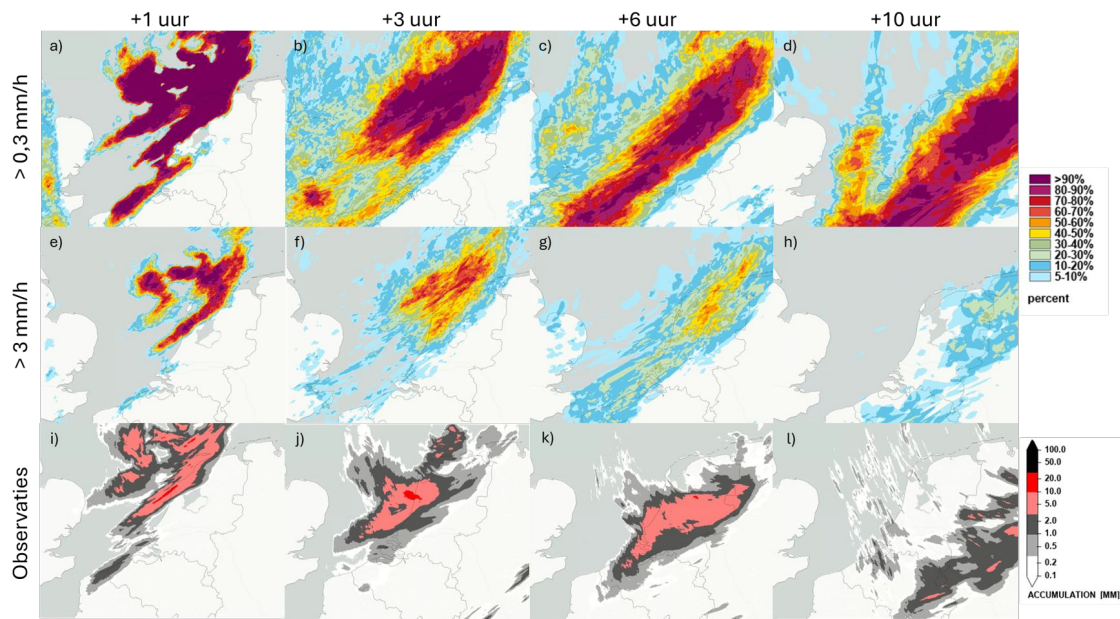
Seamless forecast (blending):

- Seamless ensemble forecast
- Issued every 5 min.
- Up to 6 hours ahead then 100% NWP.
- 20 Ensemble members from lagged NWP forecast.

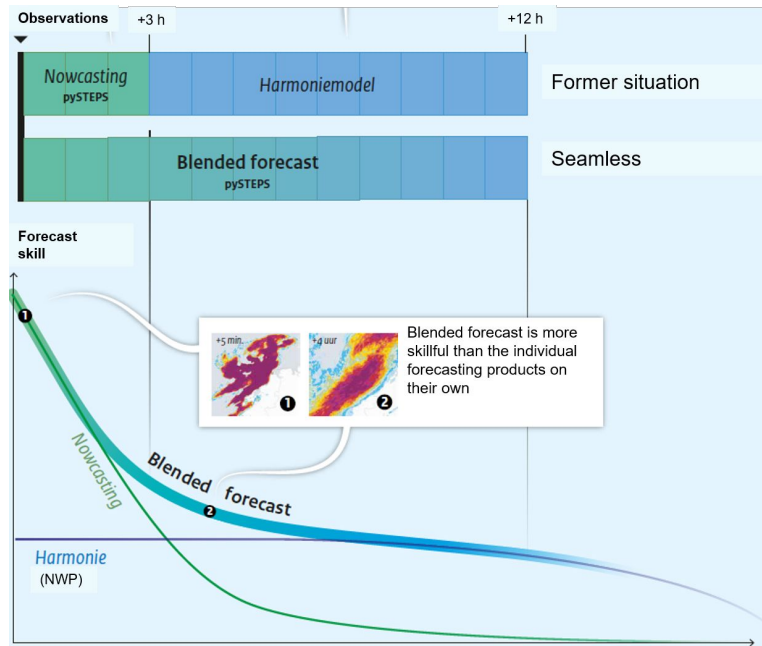
Going seamless: Operationalization of pysteps at KNMI



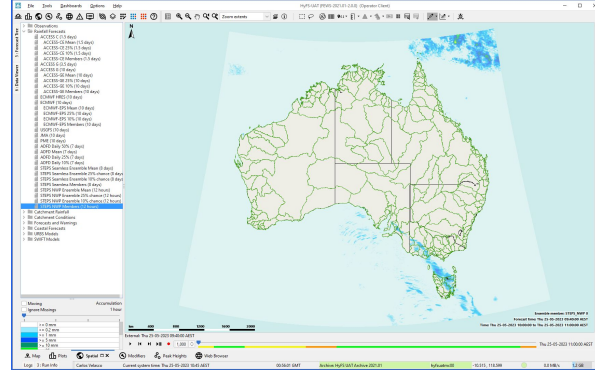
Going seamless: Operationalization of pysteps at KNMI



What's next? → NWP at 15-min time step, so better matching with nowcast in blending!

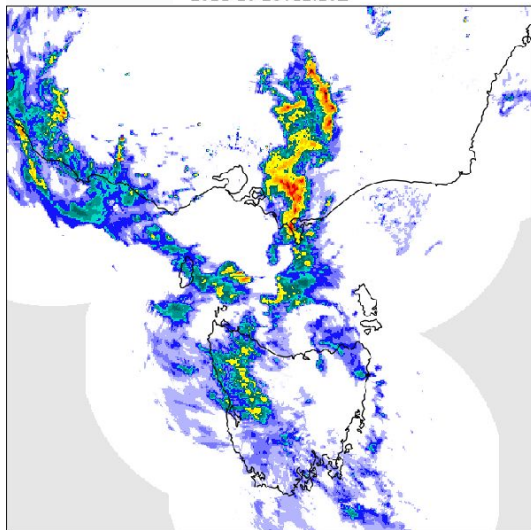


STEPS-NWP at Bureau of Meteorology, Australia



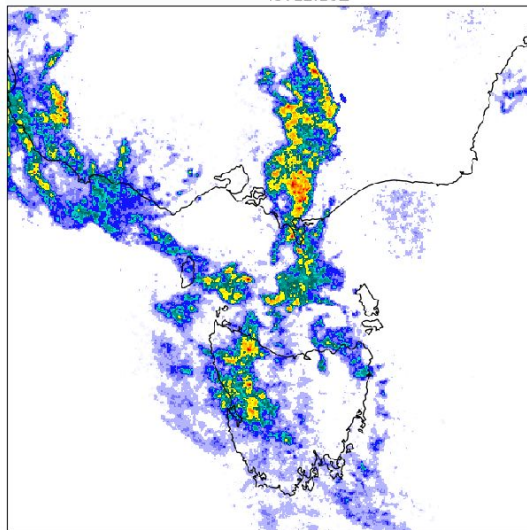
RADAR

Red radar rainfall rate (10min)
Vic-Tas domain
2021-10-28T12:10Z



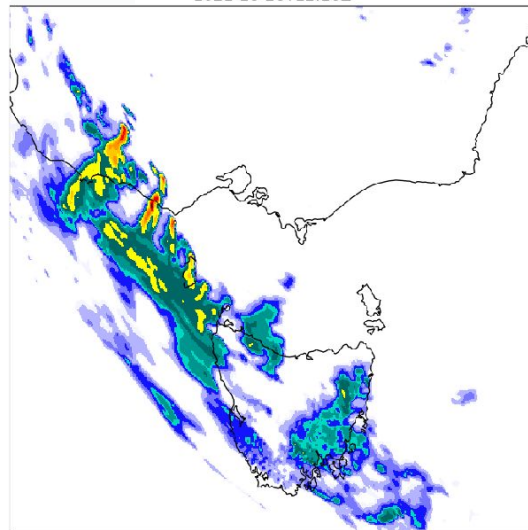
STEPS-NWP

STEPS-NWP rainfall rate
2021-10-28T12:00
adtime -10 mins
2021-10-28T12:10Z



NWP

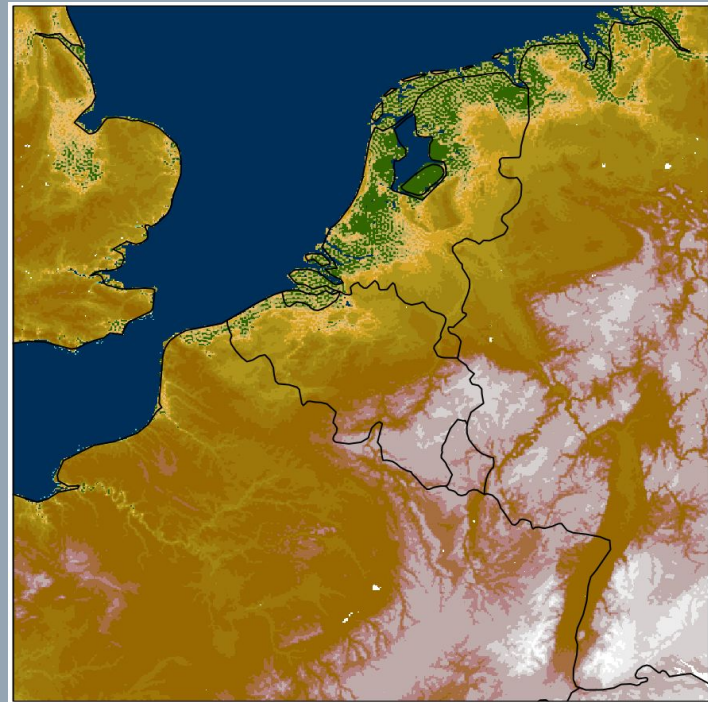
Blended G3CE3 rainfall rate
Run time : 2021-10-28T06:00
Member - 03
2021-10-28T12:10Z



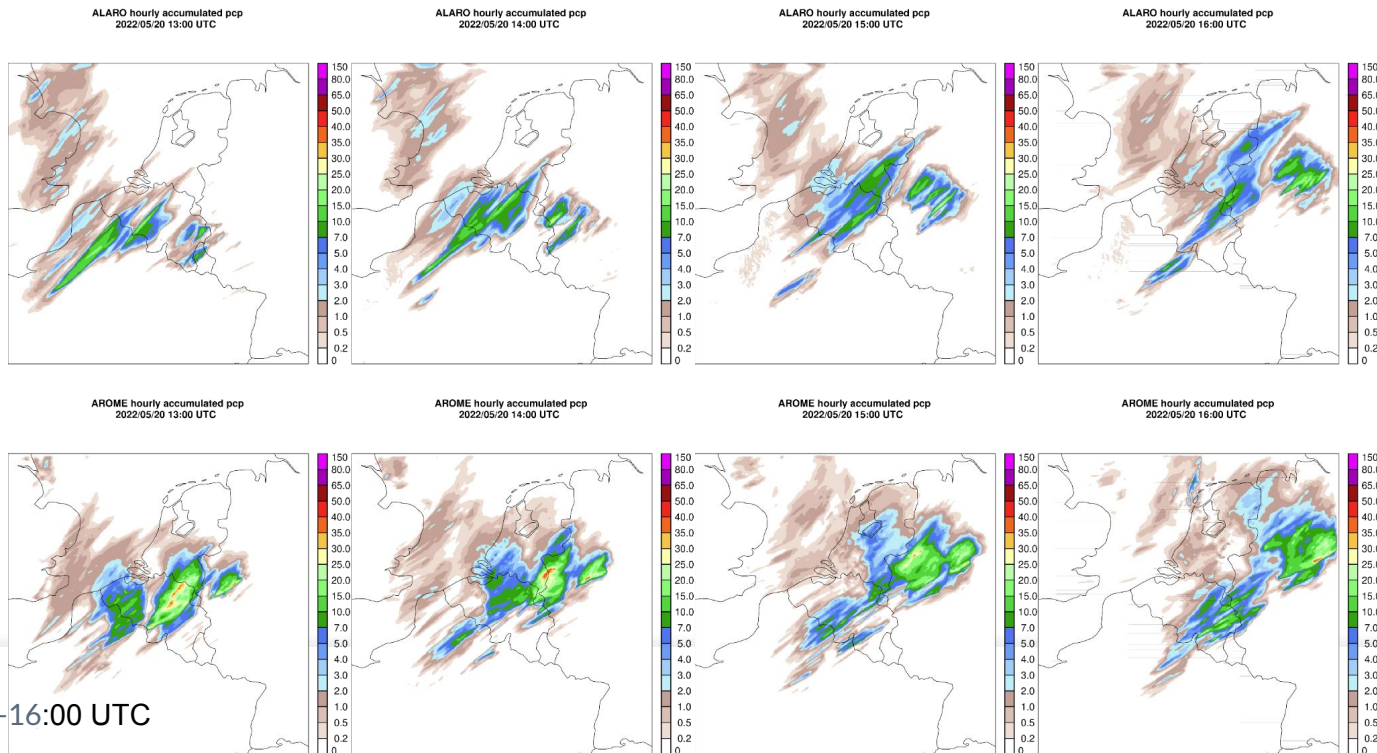
- Seamless nowcasts with 5' timesteps, next 6 hours, 1.3km resolution.
- Combines two sources:
 - Radar observations (RADQPE2)
 - Numerical Weather Prediction (ALARO/AROME 1.3km)
- STEPS blending from nowcast to NWP
- 24 ensemble members
- Runs every 15' (improvements & tuning have enabled 5' runs)

2 ACCORD MODELS: ALARO & AROME

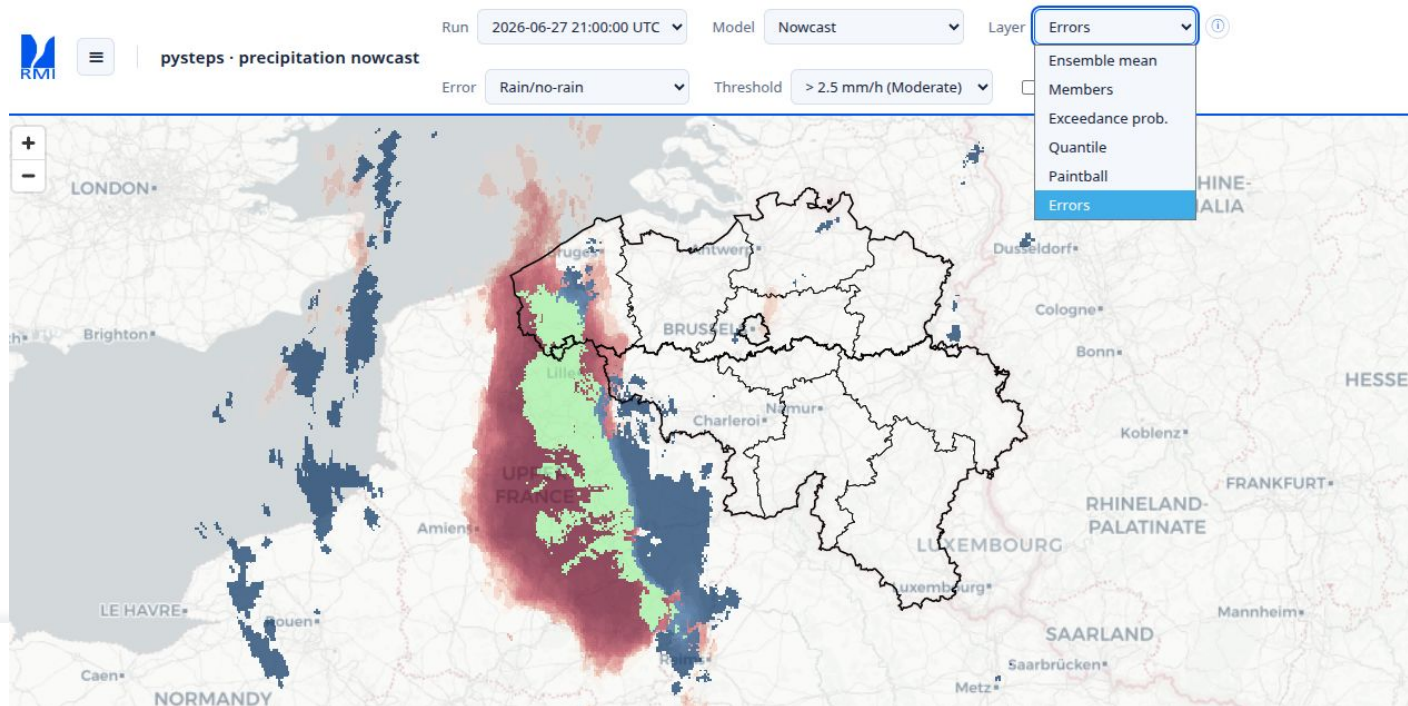
- coupled to ARPEGE (MF) | IFS (ECMWF)
- *in-house* models (= full control)
- 4 runs/day (0, 6, 12, 18 UTC)
- 3h data assimilation cycle (AROME)
- +48h forecast range
- 45s timestep
- 1.3 km horizontal resolution



- Difference for a convective case



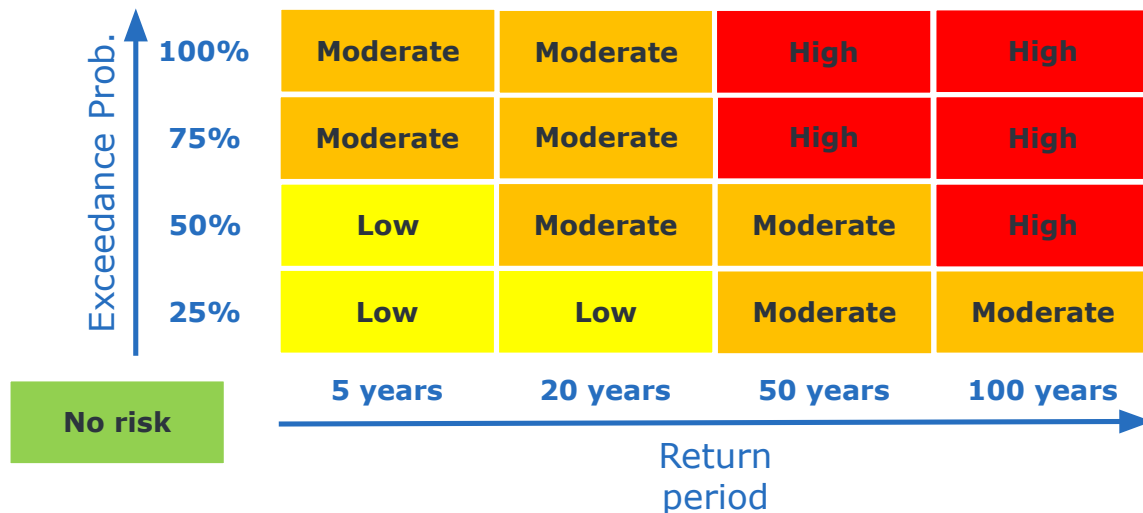
- New interface available for the weather office (by Lars Bonnefoy)
- Application: pyRainWarn probabilistic microwarnings (Felix Erdmann)



Microwarnings with pyRainWarn

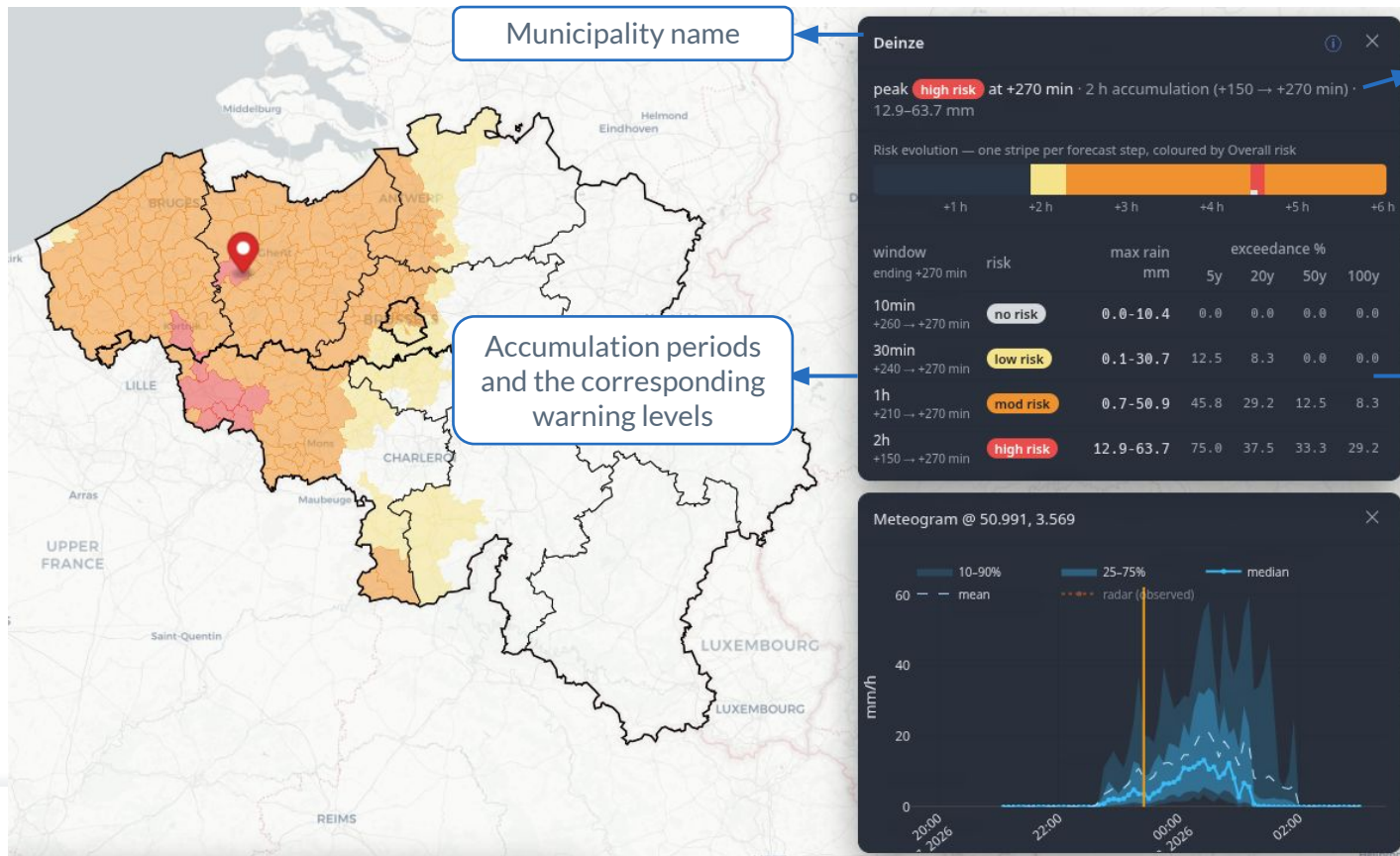
How to translate ensembles info into something easily interpretable?

- 4 colors representing the risk of extreme rainfall based on **return period** and **probability of occurrence anywhere in the municipality**.



- 4 return periods for 4 aggregation times – warning color as the most extreme result, all results stored in a table

Microwarnings with pyRainWarn



Municipality name

Accumulation periods and the corresponding warning levels

Peak warning level during all lead times and first time of occurrence

For each accumulation: Exceedance probabilities for the 4 return periods and corresponding warning levels.

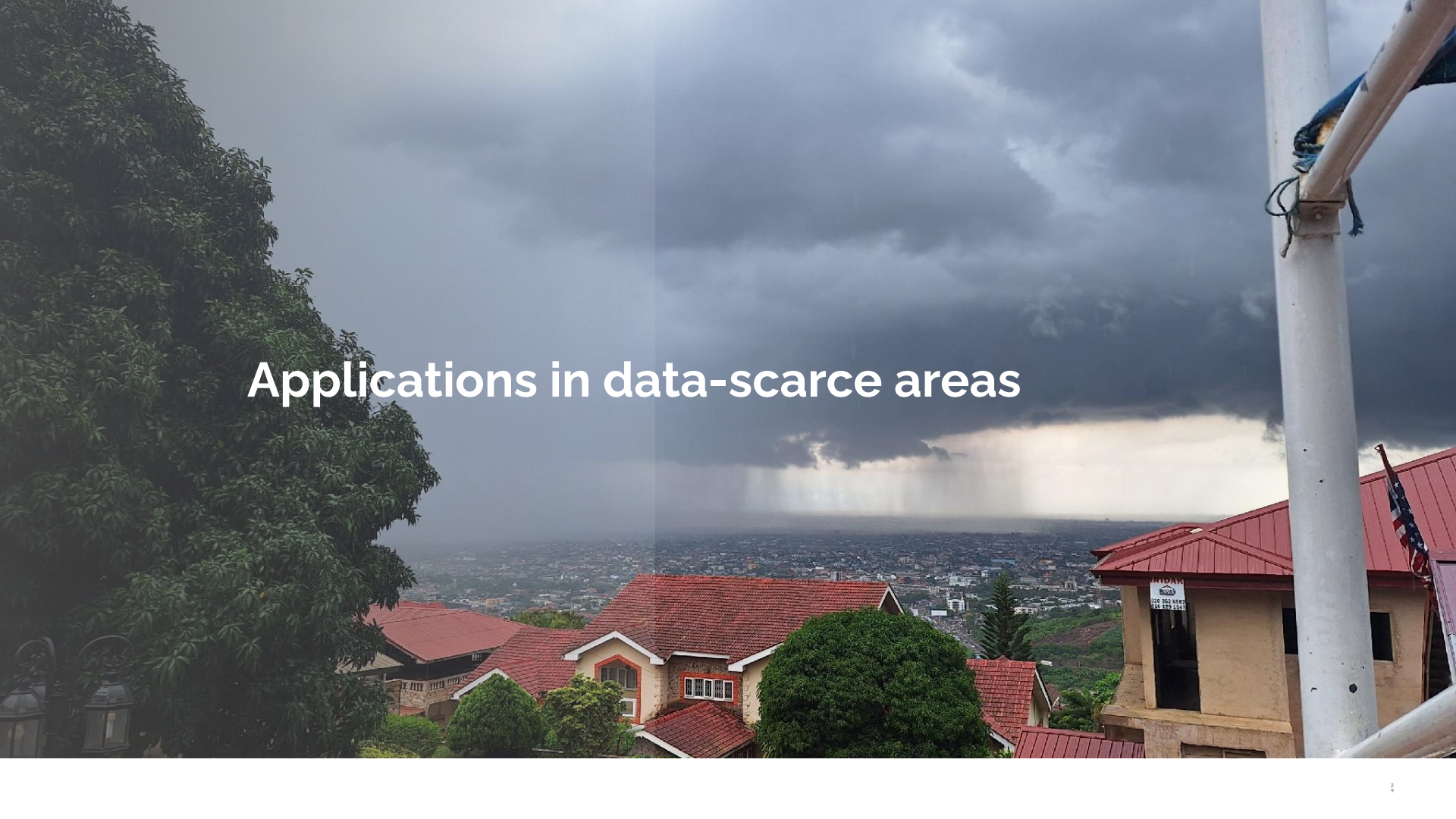
Please see the posters for more!

Monday, 13:30-15:50, Poster 146 (pyRainWarn poster):

"Advancing extreme rainfall warnings in Belgium through radar QPE-driven pySTEPS-BE ensemble nowcasting" by Felix Erdmann et al.

Monday, 13:30-15:50, Poster 240:

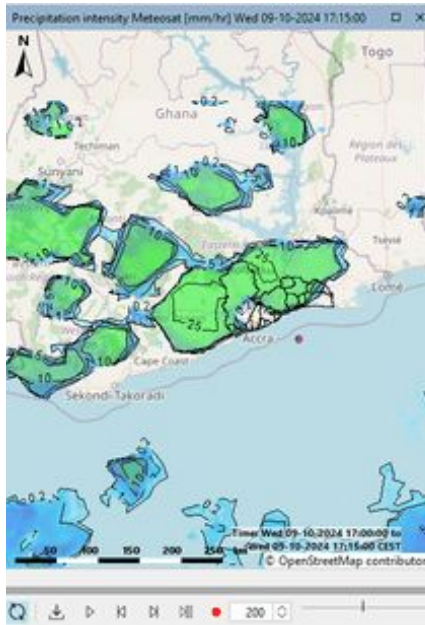
"Evaluation of the Operational Blended Precipitation Nowcasting System in Belgium (pySTEPS-BE)" by Ricardo Reinoso-Rondinel et al.

A dramatic landscape photograph showing a cityscape under a heavy, dark, stormy sky. In the foreground, there are houses with red-tiled roofs and lush green trees. A white utility pole is visible on the right side. The text "Applications in data-scarce areas" is overlaid in the center of the image.

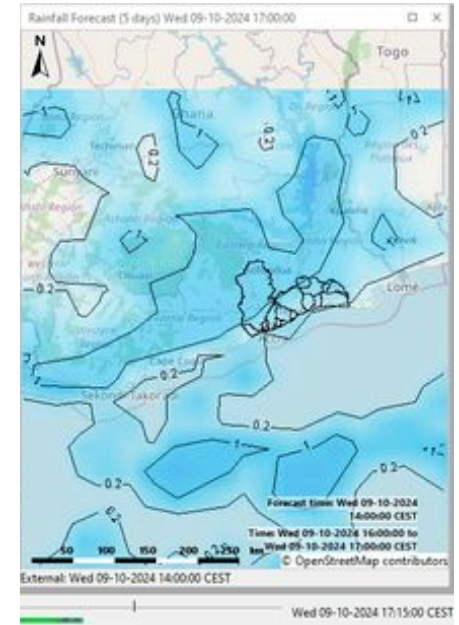
Applications in data-scarce areas

Satellite-based rainfall nowcasting as alternative

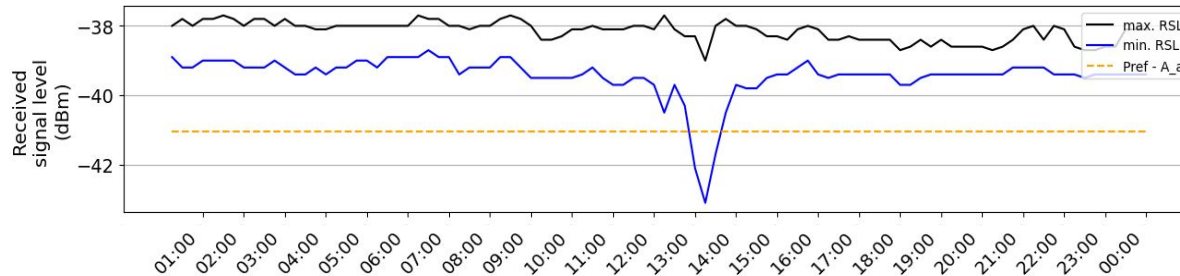
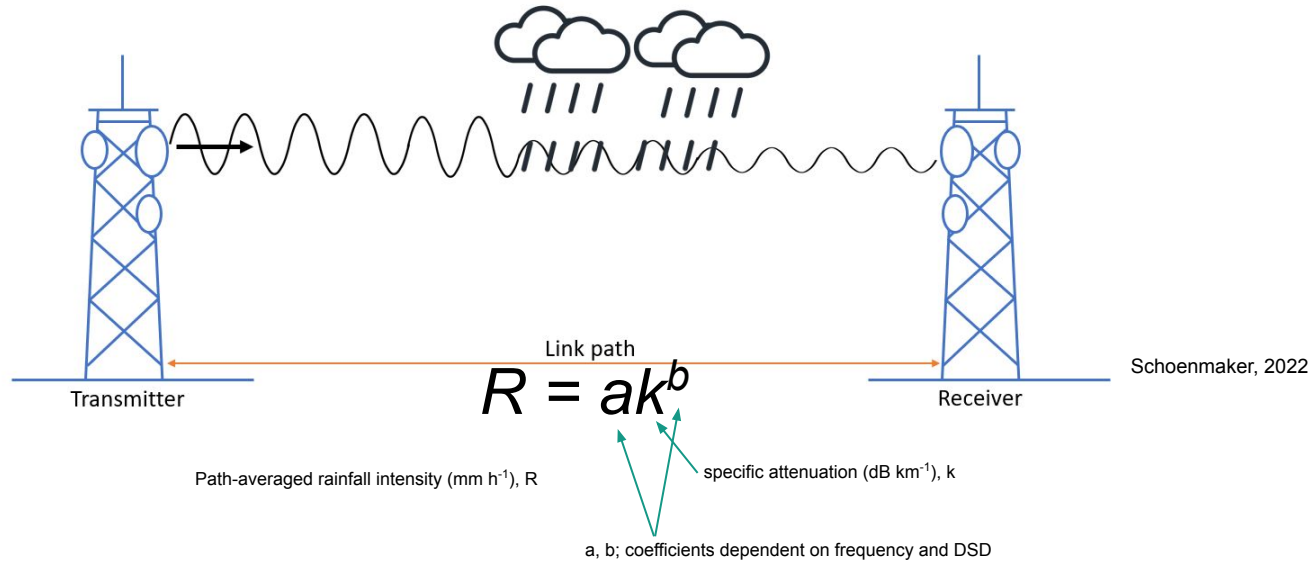
Meteosat observations STEPS



ge forecast



Rainfall forecasting with alternative sensors

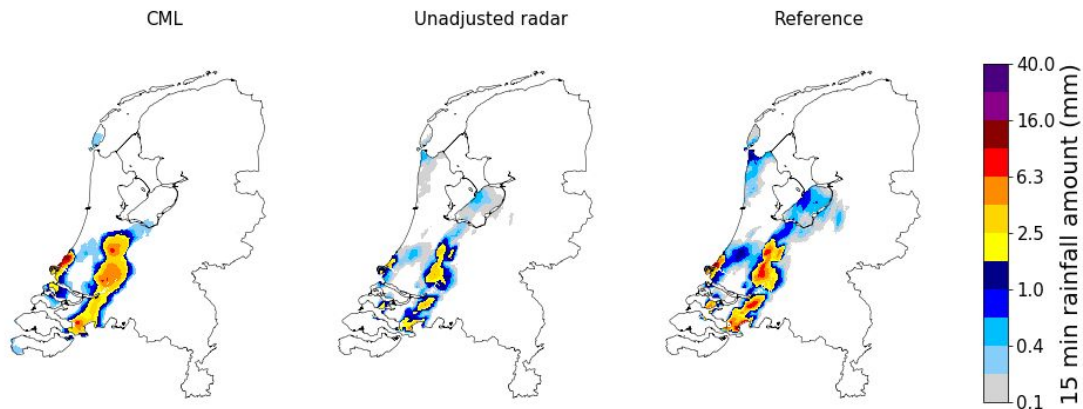


Walraven et al., 2026,
WRR (in review)

Rainfall forecasting with alternative sensors

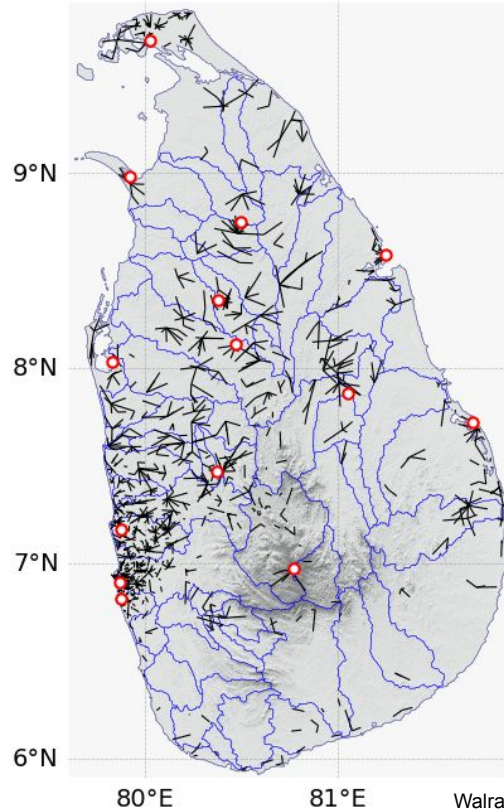
Application for the Netherlands

2011-09-10 20:00 UTC + 0 min.



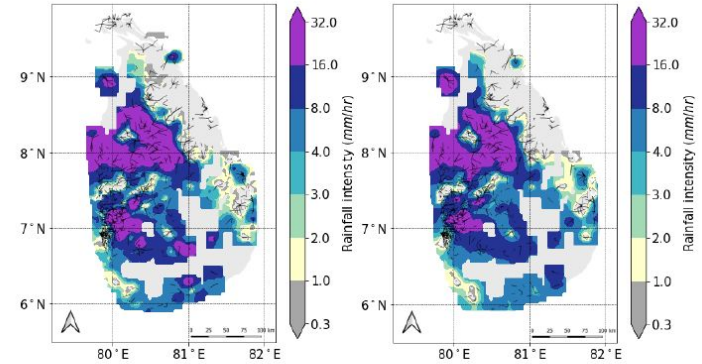
Courtesy of Bas Walraven

Rainfall forecasting with alternative sensors



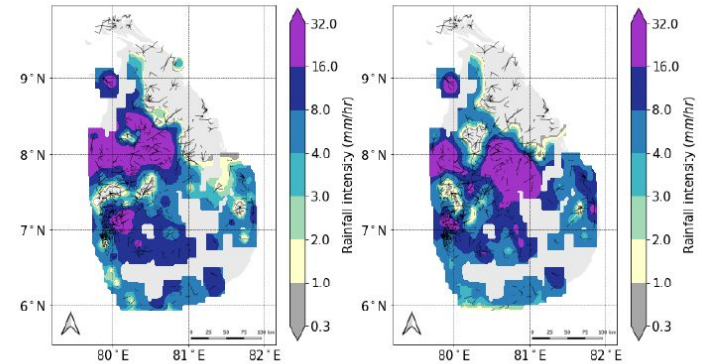
Walraven et al., 2026,
WRR (in review)

Application for Sri Lanka



(a) QPE at 6:00 pm

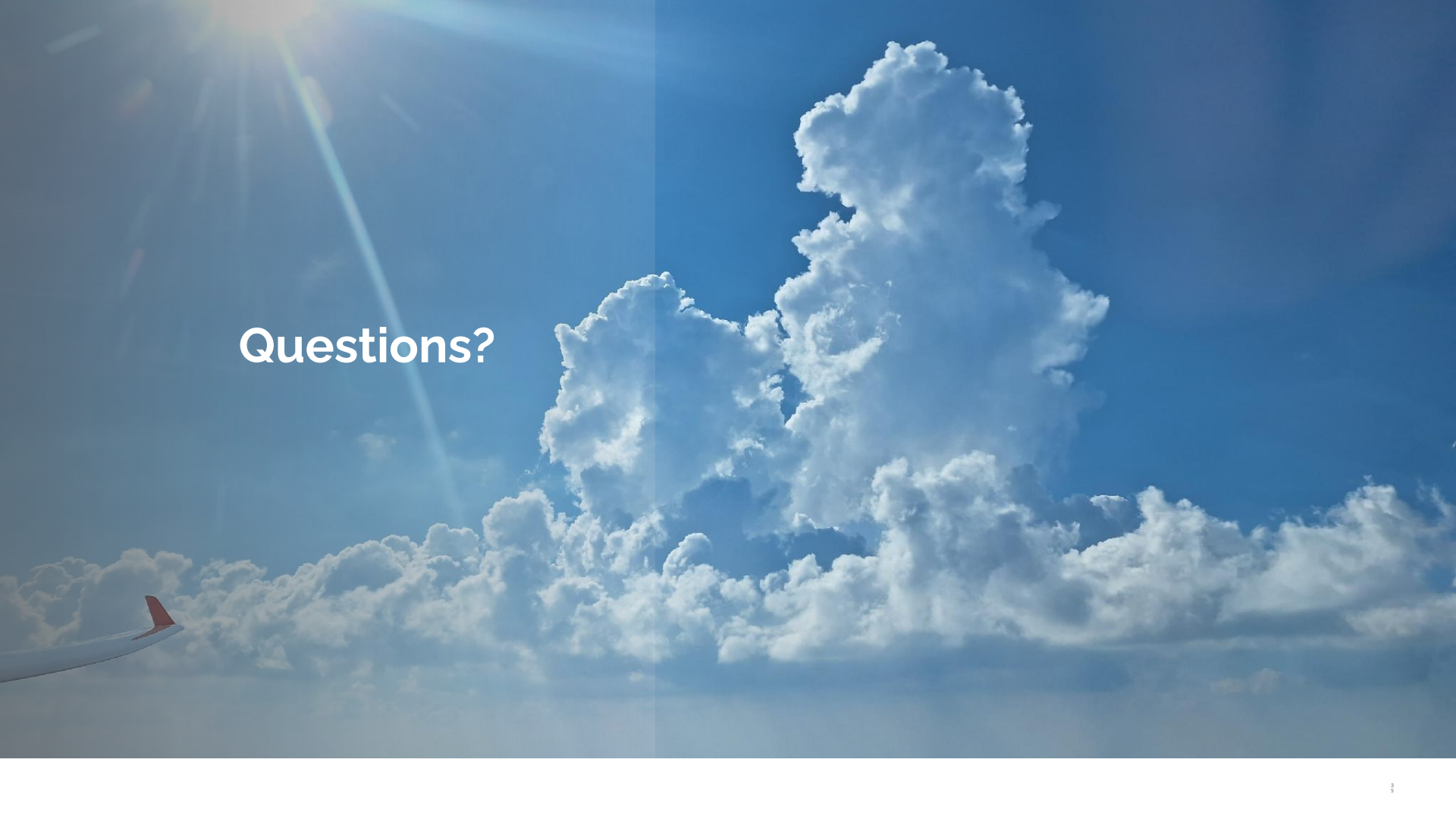
(b) Pysteps-D: 15 min lead time



(c) Pysteps-D: 30 min lead time

(d) Pysteps-D: 60 min lead time

Schoenmaker, 2022



Questions?



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